

ACTIVITY 4 — FACILITATOR GUIDE**Improve — Choose and Pilot a Countermeasure**

TYPE	DURATION	MAPS TO	TOPIC
Hands-on Activity	30 minutes	IMPROVE	Improve — Fix the Cause

Purpose of this activity

By the end of this activity learners will be able to describe improvement actions that address the identified cause (A5, K2).

What learners produce

A brainstormed countermeasure list, a completed impact/effort grid with one countermeasure selected, standard work for the new method, and a one-week pilot plan naming a measure, an owner and a review date.

Materials required

- Learner Worksheet (one per learner)
- Checklist (one per group)
- Flipchart or A3 sheet and markers
- The course slides for the scenario and framework
- Internet access where the activity calls for live research

Set-up

Groups of 3–5. Distribute one worksheet per learner and one checklist per group, and make sure every group can open this activity's data/ folder. All 5 activities run on the same continuous BrewBean Cafe scenario and follow DMAIC in order, so remind learners where this activity sits in that story — and that its output feeds the next one.

Scenario / briefing material

The team has proved its root cause and has a week to try something. The owner has agreed to a pilot but has made clear there is no budget for equipment this quarter.

Twelve countermeasures were generated at the last team meeting, ranging from adding the counter milk restock to the opening checklist through to installing a mobile pre-order app. They vary enormously in cost, effort and risk, and the team has limited time and no capital.

A cause is only worth finding if something changes because of it. Your job is to choose one countermeasure small enough to try next week, write the new method down so anyone on shift could follow it, and plan how you will know whether it worked.

How to run it — step by step

#	Instruction to learners	Facilitator note
1	Restate the root cause. Write your top root cause from Activity 3 at the top of templates/countermeasure-brainstorm-template.csv. Every idea from here on must address THIS cause. The discipline of Improve is solving the cause you proved, not the symptom you started with.	Keep data/pilot-week-results.csv HIDDEN until the pilot plans are written — revealing it early removes the thinking.
2	Brainstorm without judging. Generate at least six countermeasures without evaluating any of them yet. Then compare your list with data/countermeasure-options.csv, which holds the twelve the team generated, with impact, effort, cost and risk already scored.	Watch for tables choosing the exciting option (the pre-order app) over the effective one. Ask which proved root cause it addresses.
3	Check the Lean tools. Two quick tests. 5S: is anything at the counter hard to find, out of place or untidy? Mistake proofing (Poka-Yoke): could the wrong order be made impossible rather than merely discouraged? Both often produce cheaper fixes than adding people.	Send back any countermeasure aimed at the symptom ('hire more staff so the queue is shorter') to the Activity 3 shortlist.
4	Plot the impact/effort grid. Using the Impact and Effort scores in the data, plot every idea on templates/impact-effort-grid-template.csv and pick ONE from the high-impact, low-effort corner. Force a spread when scoring — if everything is high impact, the grid tells you nothing.	Standard work written as a paragraph of prose is not standard work. Numbered steps.
5	Write the standard work. In templates/standard-work-and-pilot-template.csv, write the improved method as short numbered steps anyone on shift could follow, ending in an unambiguous done/not-done check. If you write 'make sure there is enough milk', ask how the next person would know.	Thursday is the teaching point of the whole lab: the restock was missed and every measure snapped back. Let the room find it, then ask what would have prevented it — every answer they give is a CONTROL, and that is the agenda for Activity 5.
6	Write the pilot plan. Complete the pilot rows: what changes, who runs it, the start and end dates, what you will measure, the baseline to beat, how you will know it worked, and the review date. Pilot ONE countermeasure — change two and you will not know which one worked.	
7	Compare against what actually happened. Now open data/pilot-week-results.csv and read the week day by day against your plan. Pay close attention to Thursday.	

	Ask yourselves what Thursday tells you about the countermeasure — and about what is still missing.	
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Suggested timing

Minutes	Phase
4 min	Restate the root cause and brainstorm against it
9 min	Score and plot every idea on the impact/effort grid
6 min	Write the chosen method as standard work
7 min	Write the pilot plan, then compare with the real results
4 min	Debrief the Thursday dip

Total: 30 minutes

Discussion & decision prompts

1. Which countermeasures address the proven root cause, and which quietly address the symptom instead?
2. What does the impact/effort grid tell you that a simple list of ideas does not?
3. Why pilot only ONE countermeasure rather than several at once?
4. Looking at the pilot week results — what happened on Thursday, and what does it reveal?

Debrief

Bring the class back together and draw out the learning:

- Ask two or three groups to show what they produced, not just what they concluded.
- Ask what surprised them — the gap between what they assumed and what they found.
- Link the activity explicitly back to DMAIC · IMPROVE — Improve — Fix the Cause.
- Remind learners this activity rehearses what the Case Study assessment will require.

Reflection questions:

- Was your instinct to pick the most impressive solution or the most effective one?
- Would your standard work be followable by someone on their first day, with no explanation?
- The pilot worked on four days of five. Is that a success or a failure, and why?

Facilitator notes — what to watch for

- Keep data/pilot-week-results.csv HIDDEN until the pilot plans are written — revealing it early removes the thinking.
- Watch for tables choosing the exciting option (the pre-order app) over the effective one. Ask which proved root cause it addresses.
- Send back any countermeasure aimed at the symptom ('hire more staff so the queue is shorter') to the Activity 3 shortlist.
- Standard work written as a paragraph of prose is not standard work. Numbered steps.
- Thursday is the teaching point of the whole lab: the restock was missed and every measure snapped back. Let the room find it, then ask what would have prevented it — every answer they give is a CONTROL, and that is the agenda for Activity 5.

Success criterion

Every group can show a countermeasure that addresses the Activity 3 root cause, a completed impact/effort grid with one selection, numbered standard work with a done/not-done check, and a pilot plan naming a measure, an owner and a review date.

Assessment link

This activity builds IMPROVE — Improve — Fix the Cause. All 5 activities are in-class activities assessed indirectly through the open-book Case Study (CS) component, which reuses the same techniques on the same continuous BrewBean Cafe scenario. Knowledge is assessed in the Written Assessment (WA-SAQ). Nothing is assessed that is not practised here or taught in the slides.